CUBICLE -08

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DAY 6 – TWO IN ONE DIGITAL OSCILLOSCOPE

Date: 24-07-2025
Domain: Embedded Systems and IoT
Topic: Two in One Digital Oscilloscope [Task 3 , Task 4]

Task 3: PWM Duty Cycle Sweep

Objective:

To generate a PWM signal on an Arduino pin where the duty cycle sweeps from 0% to 100% and back continuously, and observe the waveform on a Digital Storage Oscilloscope (DSO).

Components Required:

- Arduino Uno
- USB Cable
- Jumper Wires
- Digital Storage Oscilloscope (e.g., FNIRSI-1014D)

Arduino Code:

```
int pwmPin = 9;

void setup() {
  pinMode(pwmPin, OUTPUT);
}

void loop() {
  // Increase duty cycle from 0% to 100%
  for (int duty = 0; duty <= 255; duty++) {
    analogWrite(pwmPin, duty);
```

```
    delay(10);
  }

  // Decrease duty cycle from 100% to 0%
  for (int duty = 255; duty >= 0; duty--) {
    analogWrite(pwmPin, duty);
    delay(10);
  }
}
```

Code Explanation:

- Pin 9 is a PWM-enabled pin on Arduino Uno.
- analogWrite (pwmPin, duty) sends a PWM signal with a duty cycle between 0 (0%) and 255 (100%).
- The loop:
    - Gradually increases the duty cycle from 0% to 100%.
    - Then decreases it back to 0%.
- A small delay of 10 ms between steps smoothens the transition.

Connection Setup:

- Connect Arduino Pin 9 to Oscilloscope Channel 1 probe.
- Connect Oscilloscope Ground to Arduino GND.

Oscilloscope Settings:

- Time/div: 0.5 ms/div
- Volt/div: 2 V/div
- Optionally enable:
    - Persistence mode (to view transition trails)
    - Infinite waveform memory (to track change history)

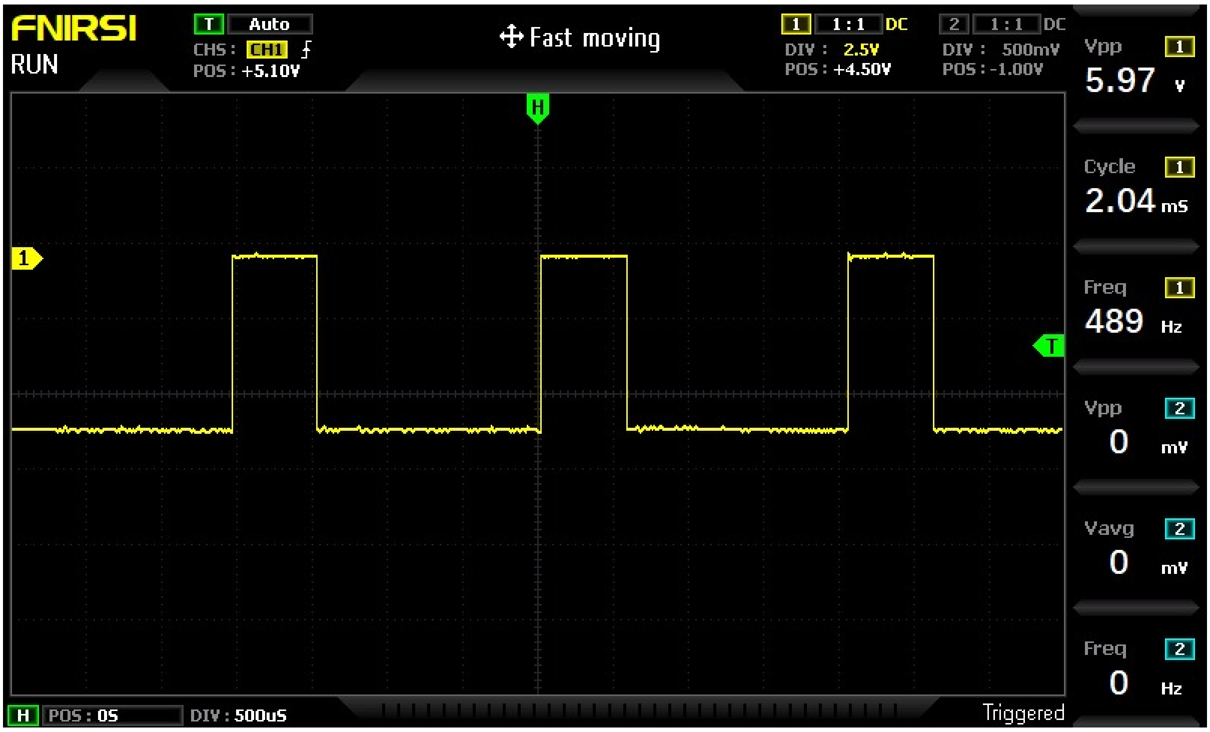
Observation:

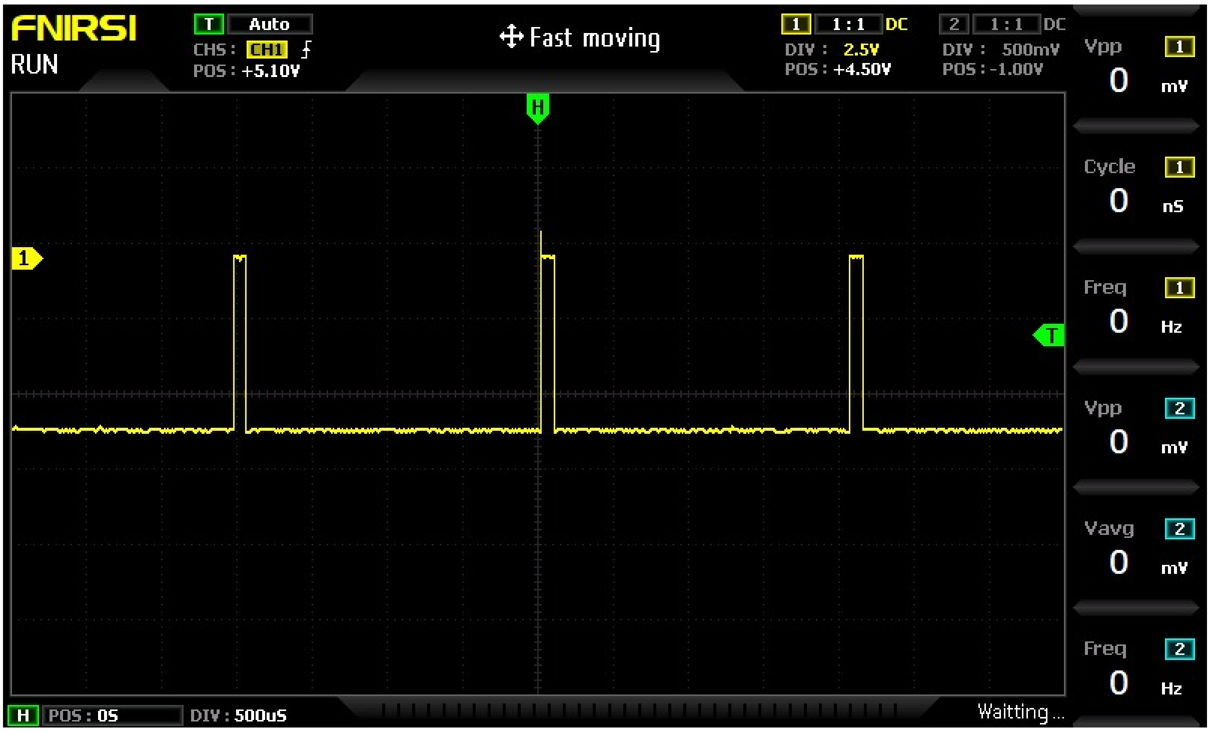
- A square waveform with a constant frequency is visible.
- The pulse width changes gradually over time.
- The duty cycle sweeps smoothly from 0% (flat line) to 100% (full HIGH) and back.
- Pulse width gets narrower and wider in a loop.

Expected Output:

- Constant frequency PWM signal
- A waveform where:
    - The duty cycle increases and decreases smoothly
    - The HIGH portion (pulse width) widens from 0% to 100% and then narrows back
- Clear visualization of PWM modulation on the DSO screen

Screenshot:





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https://drive.google.com/file/d/1YsaN-degsYyMQMtAbwfbwSh02UAPWsvE/view?usp=drive_link

Task 4: LED Dimming Using PWM

Objective:

To dim an LED using PWM (Pulse Width Modulation) and visualize the waveform controlling its brightness on a digital oscilloscope.

Components Required:

- Arduino Uno (or compatible)
- LED
- 220Ω resistor
- Connecting wires
- Breadboard
- FNIRSI-1014D Digital Oscilloscope

Arduino Code:

```
void setup() {
  pinMode(9, OUTPUT);
}

void loop() {
  // Increase brightness
  for (int duty = 0; duty <= 255; duty++) {
    analogWrite(9, duty);
    delay(10);
  }

  // Decrease brightness
  for (int duty = 255; duty >= 0; duty--) {
    analogWrite(9, duty);
```

```
    delay(10);
  }
}
```

Code Explanation:

- The analogWrite(9, duty) function generates a PWM signal on pin 9.
- The loop increases the duty cycle from 0 to 255 (0% to 100%), then decreases it.
- This causes the LED to gradually brighten and dim.

Circuit Connection:

- Connect Pin 9 of Arduino to one end of 220Ω resistor.
- Connect the other end of resistor to LED anode (long leg).
- Connect LED cathode (short leg) to GND.
- Optionally, place oscilloscope probe at Pin 9 or across LED to observe waveform.

Oscilloscope Settings (FNIRSI-1014D):

- Time/Div: 0.5 ms/div
- Volt/Div: 2 V/div
- Use Channel 1 probe connected to Pin 9 or across LED
- Use DC coupling mode
- Optionally enable Persistence Mode or Infinite Memory for visualizing pulse changes over time

Observation:

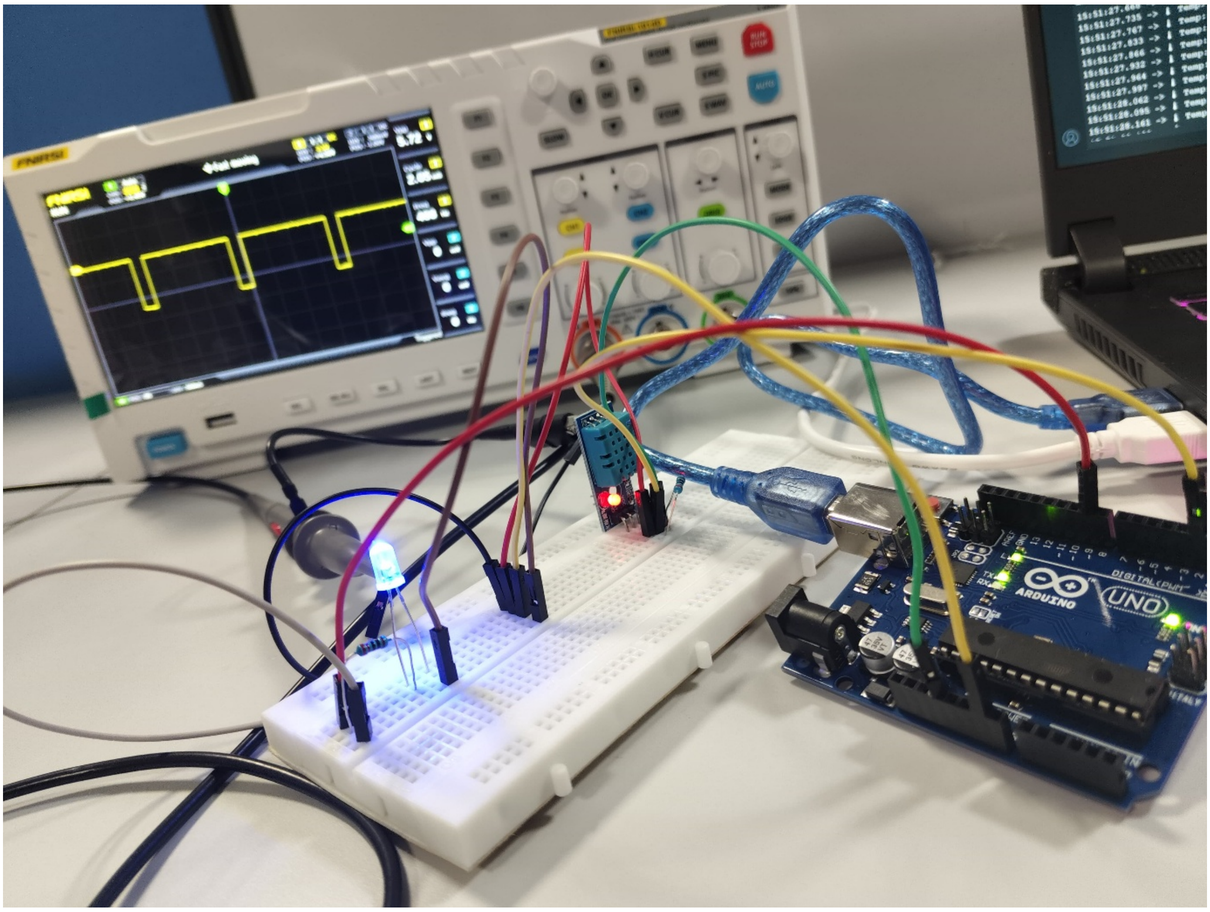
- As the duty cycle increases, the LED gets brighter.
- As it decreases, the LED dims.
- The waveform on the oscilloscope shows:

- A fixed frequency PWM signal
- Gradually increasing and decreasing pulse width
- Voltage average across LED increasing and decreasing with duty

Expected Output:

- PWM waveform with:
  - Constant frequency (~490 Hz)
  - Varying pulse width
- LED brightness gradually increases and decreases
- DSO shows the waveform becoming wider and narrower

Image:



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